SET-A

GLA University

II-Mid Term Exam, 2013-14 (Even Sem)

Soft Skill

Code- AHE-282

Course- B.Tech **Year- II** **Branch-All Branches**

Instructions :

- There are two sections in the question paper. Section I contains 25 questions and section II contains 50 questions.
- Attempt all questions.
- The maximum marks for the test are 100.
- The total time allocated is 90 minutes.
- You have to write your answers (A/B/C/D/E) in the Answer sheet.
- There will be negative marking. 25% marks will be deducted for each wrong answer.

Name of Student:-

Branch:-

Section-

University Roll No:

- **Signature of Invigilator-**

S.N.	Ans.	S.N.	Ans.	S.N.	Ans.	S.N.	Ans.	S.N.	Ans.	S.N.	Ans.	S.N.	Ans.	S.N.	Ans.
1		11		21		31		41		51		61		71	
2		12		22		32		42		52		62		72	
3		13		23		33		43		53		63		73	
4		14		24		34		44		54		64		74	
5		15		25		35		45		55		65		75	
6		16		26		36		46		56		66			
7		17		27		37		47		57		67			
8		18		28		38		48		58		68			
9		19		29		39		49		59		69			
10		20		30		40		50		60		70			

Marks in section A- **Marks in Section-B** **Total Marks (A+B)-**

Signature of Evaluator-

SECTION-I

(No of questions=25, +2 marks for each correct answer and -1/2 marks for each wrong answer)

Directions for questions (1 to 3): Select the correct alternative from the given choices.

1. What is the maximum number of identical pieces a cube can produce when cut into by 20 cuts?
   a) 441
   b) 343
   c) 512
   d) 448

2. What is the least number of cuts required to cut a cube into 504 identical pieces?
   a) 24
   b) 20
   c) 21
   d) 22

3. If A, B, C, D, E and F are written on different faces of a cube as given below showing the two different views of the cube, then which letter is written on the face opposite to D?
   a) B
   b) C
   c) E
   d) Cannot be determined

Directions for questions (4 to 5): These questions are based on the following information.
216 small but identical cubes are put together to form a large cube. This large cube is now painted on all six faces.

4. How many of the smaller cubes have no face painted at all?
   a) 125
   b) 27
   c) 64
   d) 49

5. How many of the smaller cubes have exactly two faces painted?
   a) 56
   b) 60
   c) 64
   d) 48

Directions for questions (6 to 9): A cube is divided into 1728 identical cubelets. Each cut is made parallel to some surface of the cube . But before doing that the cube is coloured with green colour on one set of adjacent faces, red on the other set of adjacent faces, blue on the third set.

6. So, how many cubelets are there which are painted with exactly one colour & on one face only?
   a) 343
   b) 216
   c) 729
   d) 600
   e) None of these

7. So, how many cubelets are there which are painted with exactly one colour & on one face only?
   a) 373
   b) 246
   c) 759
   d) 630
   e) None of these

8. How many cuts you have made?
   a) 15
   b) 30
   c) 33
   d) 27
   e) None of these

9. How many cubelets are there which are painted with exactly two colours?
   a) 48
   b) 76
   c) 51
   d) 96
   e) None of these

Directions for questions (10 to 16): Select the correct alternative from the given choices.

10. Simplify 277 + 3/8 of 432 ÷ (24 + 5 - 11)
    a) 280
    b) 286
    c) 292
    d) 298

11. If the HCF of two numbers equals to their LCM, the numbers must be
    a) Equal
    b) Relatively prime
    c) Prime
    d) None of these

12. There are some soldiers. If they arrange themselves in rows, such that there are 17 soldiers per row, 1 soldier will be left behind. If they arrange themselves in rows, such that there are 39 soldiers per row, 3 soldiers will be left behind. What is the least possible number of soldiers?
    a) 86
    b) 81
    c) 120
    d) 137

13. Find the largest number which leaves a remainder of 1 when it divides 325 or 217?
a) 108
b) 54
c) 27
d) 18

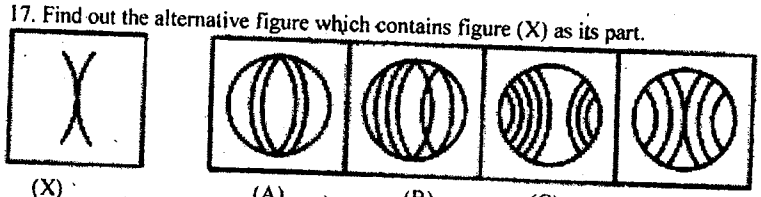
14. Find the greatest number that will divide 43, 91 and 183 so as to leave the same remainder in each case.
a) 4
b) 7
c) 9
d) 13

15. Six bells commence tolling together and toll at intervals of 2, 4, 6, 8 10 and 12 seconds respectively. In 30 minutes, how many times do they toll together?
a) 4
b) 10
c) 15
d) 16

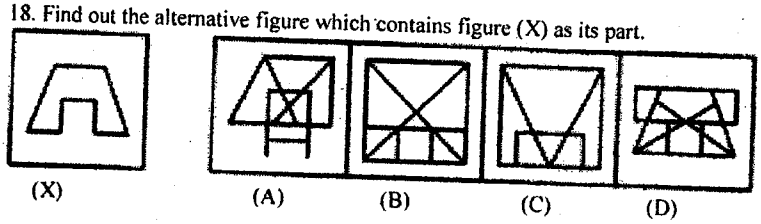
16. Let N be the greatest number that will divide 1305, 4665 and 6905, leaving the same remainder in each case. Then sum of the digits in N is:
a) 4
b) 5
c) 6
d) 8

Directions for questions (17 to 19):In each of the following questions, you are given a figure (X) followed by four alternative figures (A), (B), (C) and (D) such that figure (X) is embedded in one of them. Trace out the alternative figure which contains fig. (X) as its part.

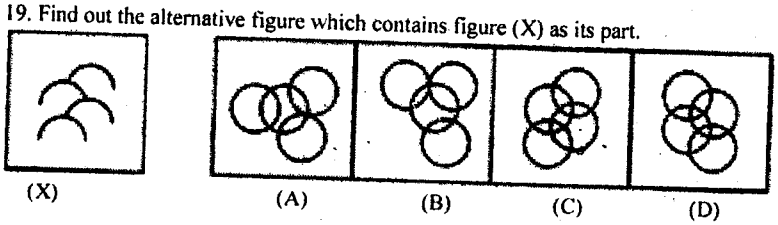
17. Find out the alternative figure which contains figure (X) as its part.



18. Find out the alternative figure which contains figure (X) as its part.

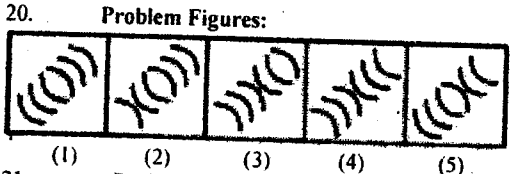


19. Find out the alternative figure which contains figure (X) as its part.

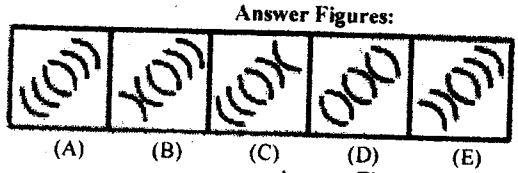


Directions for questions (20 to 25): Select a figure from amongst the Answer Figures which will continue the same series as established by the five Problem Figures.

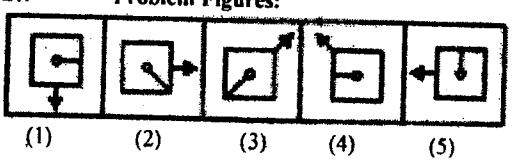
20. **Problem Figures:**



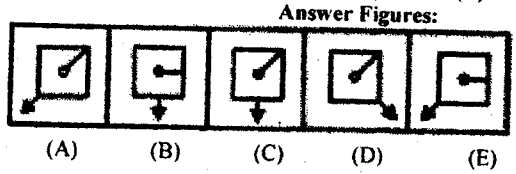
Answer Figures:



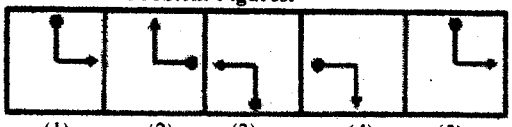
21. **Problem Figures:**



Answer Figures:

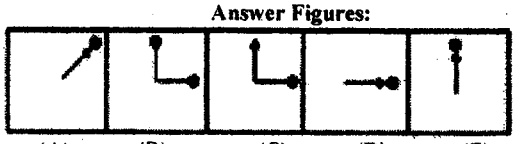


22. **Problem Figures:**



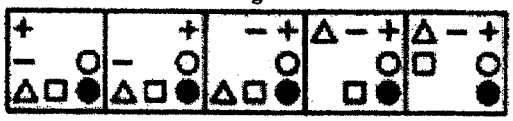
(1) (2) (3) (4) (5)

Answer Figures:



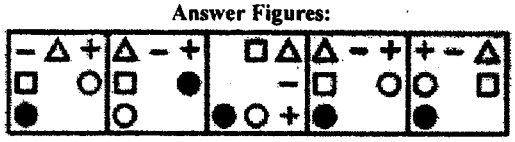
(A) (B) (C) (D) (E)

23. **Problem Figures:**



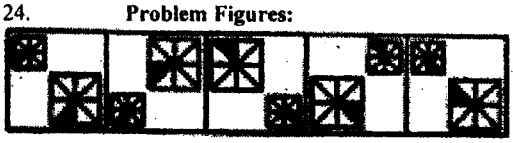
(1) (2) (3) (4) (5)

Answer Figures:



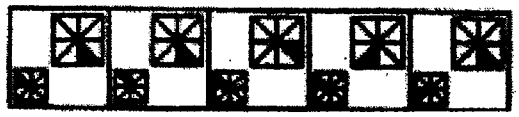
(A) (B) (C) (D) (E)

24. **Problem Figures:**



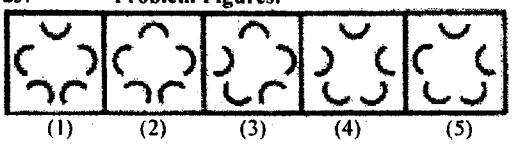
(1) (2) (3) (4) (5)

Answer Figures:



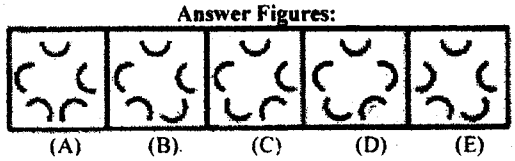
(A) (B) (C) (D) (E)

25. **Problem Figures:**



(1) (2) (3) (4) (5)

Answer Figures:



(A) (B) (C) (D) (E)

SECTION-II

(No of questions=50, +1 marks for each correct answer and -1/4 marks for each wrong answer)

26. Body language contributes_____% of how we communicate one to one.
    a) 50
    b) 55
    c) 58
    d) 53

27. _____ is an example of filters in communication.
    a) Superiority complex
    b) Cross talk
    c) Noise
    d) Language

28. Barriers to communication can be overcome by
    a) Active listening
    b) Not asking questions
    c) Making assumptions
    d) Avoid paraphrasing

29. As a professional, your communication skills have a direct influence on-
    a) Personal grooming
    b) Relationship with clients, colleagues & others
    c) Company Products
    d) Intrapersonal communication

30. The external obstacles are distortions which exist outside both the sender and receiver, and prevent effective communication. These are called-
    a) Channels
    b) Barriers
    c) Filters
    d) Means

31. Which of the following in not a barrier?
    a) Disturbance due to fan
    b) Rigid attitude
    c) Noise due to cross talking
    d) Loud Sound in theatre

32. Which of the following two are filters?
    1) Language
    2) Pre-conceived notions
    3) cross-talk
    4) disinterest

    a) 1, 3
    b) 2,4
    c) 3,4
    d) 1,4

33. Which of the following constitutes the components of communication?
    a) Verbal and Non Verbal
    b) Verbal, Non Verbal and Para Verbal
    c) Verbal and Para Verbal
    d) Non verbal and Para Verbal

34. Which of the following is not non-verbal communication?
a) Keeping your arms folded while talking
b) Leaning forward to talk to the customer
c) Making a good eye contact with the audience
d) Maintaining a good intonation pattern while talking

35. What does leaning backwards against an object indicate?
a) Closed , unreceptive
b) Feeling of negativity
c) Lack of confidence
d) Not interested

36. Which of the following is true for a positive facial expression?
a) Biting lips too frequently.
b) Blinking eyes too often.
c) Smiling while meeting somebody.
d) Raising eyebrows repeatedly.

Choose the correct option from A to D:
a) a
b) b
c) c
d) d

37. A theatre play was going on. There were certain distractions. Identify one listed as a Barrier or a Filter. Rohan's father was ill so he was thinking about his health only.
a) Barrier
b) Filter

38. Avoiding eye contact indicates_____
a) Closed, unreceptive
b) Feeling of negativity
c) Lack of confidence
d) not interested

39. Which of the following is true for positive body language?
I. It regulates the flow of communication.
II. Signals interest in others.
III. Conveys interest, concern, warmth.
IV. Conveys speaker's credibility.

Choose the correct option from A to D:
a) IV, II, I
b) III, II, I
c) IV, III, I
d) All of the above

40. Which of the following is not an example of verbal communication?
a) Teacher discussing a topic in the class.
b) Student leader giving a public speech.
c) Boss raising an eyebrow at the fault of an employee.
d) Senior officer angrily shouting at his subordinate.

41. Reema is in a meeting. She is getting upset at the continual interruption of the proceedings by another attendee. She is not the Chairperson, and is not in a position to make a comment, but she knows that she will explode if she doesn't take a break. She quietly excuses herself so she can leave the room, get a drink, and calm down before she returns. What is she demonstrating?
a) That she is submissive and can't deal with conflict.
b) That she is aware of her feelings and has a strategy to deal with the situation.
c) That she is impatient and unable to deal with the situation.
d) That she is rude, and has no appreciation of the consequence of leaving the meeting.

42. Dina is going for an interview. She doesn't believe she has any chance of securing the job, and expects to perform badly. She doesn't get the job. What has happened?
a) Dina had bad luck on the day.
b) Dina started on a negative note.
c) Dina had not prepared for the interview.
d) Dina was confused.

43. Rakesh is Preet's supervisor. He is aware that Preet is struggling at home due to the terminal illness of her partner. Preet has no official leave left. Rakesh puts himself in Preet's shoes and negotiates extended deadlines on projects, as well as allowing Preet flexible work hours and an ability to work from home on a regular basis. What is Rakesh demonstrating toward Preet?
a) Empathy
b) Sympathy
c) Social awareness
d) Frustration

44. One can increase his or her self-awareness by
a) Engaging in personal reflection and seeking feedback from others.
b) Purchasing self-help magazines.
c) Ignoring that it exists.
d) Trying to be more like those he or she admires.

45. When I make a mistake it helps me to think that -
a) I can't do anything right.
b) Now I know what not to do.
c) Everyone makes mistakes sometimes.
d) I'll get it right from now on.

46. Which of the following constitutes interpersonal skills?
a) Self Awareness
b) Assertiveness
c) Self Confidence
d) Team Building

47. When you are assertive, you are actually
a) Being violent
b) Being forceful
c) Aiming for a win-win situation
d) Making others feel good

48. _____ reflects a person's overall evaluation or appraisal of her or his own worth.
a) Self Awareness
b) Self Esteem
c) Social comparison
d) Self Image

49._____is an attitude which allows individuals to have positive yet realistic views of themselves
a) Self Awareness
b) Self Esteem
c) Self Confidence
d) Self Image

50. Self awareness in a person implies-
a) The ability to learn from your mistakes.
b) Knowing most of your strengths and weaknesses.
c) Taking a risk in grey areas.
d) Using past experience for any action.

51. Interaction between 3-12 people, who share a common goal and who attempt to influence one another is known as:
a) Personal Communication
b) Business Communication
c) Group Discussion
d) Debate

52. When a group agrees to support and commit to the decision of the group, they have reached:
a) A solution
b) A consensus
c) An analysis
d) A conclusion

53. When participating in a group discussion, it is important to:
a) Speak
b) Argue
c) Shout
d) Agree

54. It is important to have a GD because:
a) You get to make friends.
b) It enhances your technical knowledge.
c) It helps in sharing of ideas
d) None of the above

55. Group Discussion is:
a) Proving your point to the panel.
b) Exchange of thoughts and ideas amongst people.
c) Supporting others.
d) Agreeing to what others have to say.

56. Group Discussion can be compared to a:
a) Cricket match
b) Table tennis match
c) Football match
d) None of the above

57. What kind of skills are judged in a GD?
a) Interpersonal and Intrapersonal Skills
b) Communication skills.
c) Body language
d) All of the above

58. Which of the following phrases one should use to become a part of the discussion in a GD?
a) I would like to add
b) I totally disagree with you
c) Please excuse me but this is funny
d) I want to tell you

59. Group Discussion is an important tool to judge your:
a) Grammar and Vocabulary
b) Confidence
c) Listening skills
d) All of above

60. Which of the statement is incorrect?
a) One should keep calm and quiet in a GD.
b) GD is a trait where a person is judged on the basis of his attitude and personality traits.
c) Use simple English.
d) Be assertive in a GD.

61. Which of the following quality is of utmost importance in a Group Discussion?
a) Aggressiveness
b) Assertiveness
c) Not losing temper
d) b and c

62. Which of the following does not hold true in a Group Discussion (GD)?
a) Body language is an important element
b) One should make a point strongly
c) Flexibility is appreciated
d) Facts can be altered as per the situation.

63. Giving facts and figures is a good way to initiate a GD:
a) True
b) False
c) Maybe
d) Sometimes

64. Which of the following qualities is best showcased in a GD?
a) Team work
b) Knowledge
c) Body language
d) All of the above

65. Arrange the following stages of team formation:
1) Storming
2) Adjourning
3) Norming
4) Forming

a) 4,1,3,2
b) 4,3,2,1
c) 3,2,4,1
d) 1,2,3,4

66. Which of the following stages include resolving differences and appreciating others?
a) Storming
b) Norming
c) Adjourning
d) none of the above

67. Which of the following element does not exist in a group?
a) Discussion
b) Coordination
c) Analysis
d) Estimation

68. Which of the below is an interpersonal Skill?
a) Self confidence
b) Assertiveness
c) Co operation
d) Self esteem

69. Which of the following statement is correct?
a) In a group, people might or might not have a common goal.
b) In a team, people co-operate with each other to work for a common objective.
c) A task cannot be accomplished without co ordination
d) All of the above

70. Which of the below mentioned issue is a common problem in a team?
a) Lack of confidence
b) Ego clash
c) Easy agreement with each other
d) Co-operative people

71. Which one of the following characteristics defines a team?
a) Interdependent effort
b) Independent behaviour
c) Individual goals
d) Unstable relationships

72. Work groups of employees and supervisors that meet regularly to discuss their quality problems and recommend solutions is a form of participative management called:
a) Goal oriented teams
b) Evaluation teams
c) Coordinated teams
d) Group.

73. In the second stage of group development:
a) the job task is performed.
b) intragroup conflict often occurs
c) group objectives are clearly formulated
d) close relationships are developed.

74. After which stage of a group's development has the group formed a common set of expectations?
a) Forming
b) Storming
c) Norming
d) Adjourning

75. How many people must be there to make a team?
a) 2 or more
b) 8-10
c) 5 or more
d) 1

P.T.O

II Mid Term Examination, 2013-14

Title of Paper: Environmental Studies

Paper Code: CEE-201

Time: 90 Min.

Max. Marks: 20

Section-A

Note: Attempt All Questions

1 x 5 = 5 Marks

(I) Define the term effluent.
(II) What do you mean by e-waste?
(III) Write component of solar energy.
(IV) Write four example of green energy.
(V) Explain ex-situ conservation.

Section-B

Note: Attempt Any Three Questions

2 x 3 = 6 Marks

(I) Write the effects of ozone layer depletion on human life.
(II) Explain age-sex (population) pyramid of India with neat and clean diagram.
(III) What are the objectives of solid waste management?
(IV) What is acid rain? How does it affect on Taj Mahal?

Section-C

Note: Attempt any three questions

3 x 3 = 9 Marks

(I) What is soil pollution? Describe its effects and remedies.
(II) Discuss the ill effects of modern agriculture on environment.
(III) Describe merits and limitations of hydrogen energy.
(IV) Discuss briefly international and national efforts in the conservation of biodiversity.

4. For an 8085 based system, let the following two instructions are carried out

   LXI SP, 0000 H
   PUSH D

   with D = 09 H and E = FA H assumed. Show the stack contents after PUSH operation. Also write an ASP to exchange the contents of DE register pair with that of HL register pair, using PUSH, POP instructions.

Section C

Note: Attempt any Three Questions **(3x3=9)**

1. Explain the Architecture of 8086 with the help of diagram.
2. Explain the different modes of operation of 8086 with the help of its pin diagram.
3. A list of 50 numbers is stored in memory, starting at 6000H. Find number of negative, zero and positive numbers from this list and store these results in memory locations 7000H, 7001H, and 7002H respectively
4. Define the Addressing modes of 8086 with the help of suitable example.

II

Uni. Roll No:.....

II Mid Term Examination, 2013-14
Course: B.Tech. (CSE), Year: II, Semester: IV
Introduction to Microprocessors (CSE 222)

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

1. How many control flags are there in 8086? Give their name.
2. Discuss the two pins (a) DT/ R and (b) DEN .
3. Discuss LOCK signal and TEST signal in 8086.
4. Differentiate between CALL and PUSH instructions.
5. Is direct memory to memory data transfer possible in 8086? Justify your answer.

Section B

Note: Attempt any Three Questions **(2x3=6)**

1. What is Sub routine? Explain different parameter passing methods with example.
2. What is Maskable and Non-Maskable interrupt in 8085 microprocessor? And explain the bit pattern of RIM & SIM Instruction with example.
3. Write an assembly language program to separate even numbers from the given list of 50 numbers and store them in the another list starting from 2300H. Assume starting address of 50 number list is 2200H

Uni. Roll No: .....

Second Mid Term Examination, 2013-14
Course: B.Tech. (CSE), Year: 2nd, Semester: IV

Operating Systems (CSE 221)

Time: 90 Min. **Total Marks: 20**

Notes:
1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions (1x5=5)

I. What is the need of Medium Term Schedulers?
II. What is Race Condition?
III. What is Busy Waiting?
IV. Define Semaphores?
V. Define Safe State.

Section B

Note: Attempt any Three Questions (2x3=6)

I. Explain various performance criteria of CPU Scheduling algorithms.
II. Explain Multi level Queue and Multi level feedback queue CPU scheduling.
III. Explain Peterson's Solution to Critical Section Problem and state how it satisfies all the three requirements.
IV. Define Deadlock. Also, explain the necessary conditions for a deadlock.

Section C

Note: Attempt any Three Questions (3x3=9)

I. Consider the following set of processes, with the length of the CPU-burst time given in milliseconds:

Process	Burst Time	Priority
P1	10	3
P2	1	1
P3	2	3
P4	1	4
P5	5	2

The processes are assumed to have arrived in the order P1, P2, P3, P4, P5, all at time 0.

Compute the waiting time and turnaround time for each process by drawing Gantt Chart using FCFS, SJF, a non preemptive priority (a smaller priority number implies a higher priority) scheduling.

II. Explain Reader Writers problem with its semaphore solution and specify how it meets all three requirements of solution to its critical section problem?
III. Describe all the methods for deadlock prevention.
IV. It is proposed to use a multi-resource Banker's Algorithm in an OS containing 3 resource classes. The current resource state is as given below:

Process	Allocation			Maximum			Available		
	R1	R2	R3	R1	R2	R3	R1	R2	R3
P1	2	2	3	3	6	8	7	7	10
P2	2	0	3	4	3	3			
P3	1	2	4	3	4	4			

c. Is the current allocation state safe?
d. Would the following requests be granted in the current state-
   - P1 requests (1,1,0)
   - P2 requests (0,1,0)

Second Term Examination, 2013-14
Course: B. Tech. (CSE), Year: II, Semester: IV

Computer Networks (CSE224)

Uni. Roll No:
Time: 1.5 Hrs. **Total Marks: 20**

Notes:
1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions (1x5=5)

I. What do you mean by multicasting?

II. Assuming even parity, find the parity bit for each of the following data units.
   a. 1001011
   b. 0001100

III. Find the range of addresses in the block 123.56.77.32/29

IV. We have a pure ALOHA network with 100 stations. If $T_{fr} = 1\mu s$, what is the number of frames/s each station can send to achieve the maximum efficiency.

V. A sender sends a series of packets to the same destination using 5-bit sequence numbers. If the sequence number starts with 0, what is the sequence number after sending 100 packets?

Section B

Note: Attempt any Three Questions (2x3=6)

I. Given the dataword 1010011110 and the divisor 10111,
   a. Show the generation of the codeword at the sender site (using binary division).
   b. Show the checking of the codeword at the receiver site (assume no error).

II. An Ethernet MAC sublayer receives 1510 bytes of data from the upper layer. Can the data be encapsulated in one frame? If not, how

many frames need to be sent? What is the size of the data in each frame?

III.
   a. What are the differences between classful addressing and classless addressing in IPv4? Explain with examples.
   b. What is a mask in IPv4 addressing? What is a default mask in IPv4 addressing?

IV. An organization is granted the block 16.0.0.0/8. The administrator wants to create 500 fixed-length subnets.
   a. Find the subnet mask.
   b. Find the number of addresses in each subnet.
   c. Find the first and last addresses in subnet 1.
   d. Find the first and last addresses in subnet 500.

Section C

Note: Attempt any Three Questions (3x3=9)

I. Explain in detail the IPv4 datagram fields mentioning the usage of each field.

II. Discuss in brief the different types (at least two from each type) of Multiple Access protocols.

III. Explain in detail the different flow control mechanisms adapted by the data link layer for noisy channels.

IV. Explain in brief the error detection and correction mechanisms followed by the data link layer.

Univ. Roll No:.....

II- Mid Term Examination (Even Semester), 2013-14
Course: B.Tech. (CSE), Year: II, Semester: IV

Computer Graphics (CSE-223)

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. What is transformation? Explain the need of transformation with example.

II. Write 3-D rotation matrix about y-axis.

III. What is shearing? Write a Difference between uniform scaling and differential scaling.

IV. Describe Blobby objects with example.

V. Write 2x2 rotation matrix for counter clock wise by angle π.

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. Differentiate between Parallel and Perspective Projection (At least four).

II. Drive a complete matrix for scaling about any arbitrary point P(h,k).

III. Explain Bezier curves with example. Write the properties of Bezier curves.

IV. Prove that simultaneous shearing in both directions (x and y) is not equal to the composition of pure shear along x-axis followed by pure shear along y-axis.

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. A unit cube is projected on to the xy plane find out the coordinate of the projected image using perspective projection, if center of projection is (0, 0, 10).

II. Find the coordinates of a pyramid whose coordinates are A(0,0,0), B(1,0,0), C(0,1,0) and D(0,0,1) after mirror reflection with respect to the plane passing through the origin and having normal vector whose direction is V=i+j+k.

III. Reflect diamond- shaped polygon whose vertices are A(-1,0), B(0,-2),C(0,1) and D(0,2) about horizontal line y=2.

IV. Write the significance of continuity conditions in graphics. Discuss parametric continuity conditions in detail and also differentiate it with geometric continuity condition.

Uni. Roll No: _____

Second Term Examination, 2013-14

Course: B.Tech. (CSE), Year: 2nd, Semester: 4th

Database Management System (ITE 221)

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. Explain Update Anomaly with example.
II. What are Armstrong Axioms?
III. What is Functional Dependency? Define Partial Functional Dependency.
IV. Write syntax for the referential constraint.
V. Give the order in which the following clauses are used in a Query: WHERE, HAVING, ORDER BY, GROUP BY, FROM

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. For the given schema, solve the following queries:

Sailors (sid , sname, rating , age)
Boats (bid, bname, color)
Reserves (sid, bid, day)

(a) Find sid's of sailors who've reserved a red or a green boat
(b) Find names of sailors who've not reserved boat #103(using nested query)

II. What are DEMERITS of normalization (any 2)? Define 3NF with example.

III. Consider the following relation R(A,B,C,D,E) having FDs

$D \rightarrow C, \quad CE \rightarrow A, \quad D \rightarrow A, \quad AE \rightarrow D$

Find the all possible candidate keys.

IV. Consider the following schema:

```
Recipe (rID, rName, cuisine, direction)
```

a. Find cuisine name which is duplicate.
b. List rName in descending order.

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. Explain Multi-valued Dependencies and 4NF with help of example.
II. Explain Cursors and the steps involved while using a cursor. What are different types of cursor?
III. Consider the following relation R(A,B,C,D,E,F)
Having following FDs

$A \rightarrow CF, \quad C \rightarrow D, \quad B \rightarrow E$

a. List all candidate keys of R
b. What normal form is R following, normalize it up to 3NF.

IV. Consider the following scenario and write the SQL commands to create the following table:

ItemInfo

Colum name	Data type	Description	Constraints
ItemId	Varchar2(6)	Unique Id provided for each item. (eg STN001 for stationery items)	PRIMARY KEY
ItemName	Varchar2(30)	Name of the item	NOT NULL
QtyOnHand	Number(3)	Current availability of item in the shop	NOT NULL
UnitPrice	Number(6,2)	Sell price of item per unit	Greater than 0
Class	Char(1)	Depending on the UnitPrice, items belongs to various Classes. eg: A,B,C etc.	Class of Item is either 'A' or 'B' or 'C'. (Table level constraint)
UnitOfMeasurement	Varchar2(12)	Unit used to measure the quantity (eg Kilogram, dozen etc)	NOT NULL